Pinpoint Test

Does the odometry computer measure where the robot really went? (issues #6 and #21, decision 005)

Names		Date	
Battery voltage at start		Battery at end	

OpMode: Pinpoint Test (Tests group on the Driver Station). **Needs in the config:** `pinpoint` (goBILDA Pinpoint, I2C) with both pods plugged in

Hidden by default (052): delete `@Disabled` in its file, build, deploy. **Safety:** space around the robot, hands clear, one person on STOP. Nothing moves in INIT.

Controls (gamepad 1)

Button	Does
X	reset position to 0, 0, 0° (do this before EVERY run)
B	reset AND recalibrate the IMU. The robot must be STILL for a second
A	straight-line assist on / off (starts ON, like Main TeleOp): a stick pointed almost straight drives exactly straight
Dpad up / down	speed dial: raise / lower the driving speed (starts at 50%)
Sticks	drive (bumper = slow, trigger = turbo, times the speed dial). Motors BRAKE when the sticks are released, so the robot stops where you let go. It is harder to push by hand: push slowly, or drive the moves

Setup: tape a start line and a 48 in mark on the floor. Our pods are 4-bar pods (32 mm wheels), already set in the code. Directions: +X = forward, +Y = left, +heading = counter-clockwise (decision 007).

Steps, in order

#	Do this	Expected	Actual (what happened)	If wrong
1	INIT, robot still	"Pinpoint READY" before START		stays not ready: check the I2C cable and the pods
2	X. Push or drive the robot straight FORWARD 48 in	X about +48, Y about 0		X goes DOWN: `X_POD_DIRECTION = REVERSED`. X far from 48: wrong `POD_TYPE`
3	X. Push or drive the robot straight LEFT 48 in	Y about +48, X about 0		Y goes DOWN: `Y_POD_DIRECTION = REVERSED`
4	X. Turn the robot in place exactly 1 turn counter-clockwise (tape mark)	heading back to about 0 (or 360), X and Y stay within 1 in		X or Y drift a lot: the pod OFFSETS are off, see step 5

#	Do this	Expected	Actual (what happened)	If wrong
5	X. Spin 10 turns in place, slowly	X and Y still within 2 in of 0		measure from the WHEELBASE center to each pod wheel (mm) and set `X_POD_OFFSET_MM` / `Y_POD_OFFSET_MM`
6	X. Drive a 48 in square and come back to the start	X, Y within 2 in, heading within 3°		repeat 3 times and write the worst error

Measurements

Run	What	Pushed / turned	Pinpoint read X	Y	Heading	Error
2	forward 48 in	48 in				
3	left 48 in	48 in				
4	1 turn	360°				
5	10 turns	3600°				
6a	square					
6b	square					
6c	square					

Numbers to put into the code: `subsystems/Odometry.java`

Constant	Was	New value	Checked by
`X_POD_DIRECTION`	FORWARD		
`Y_POD_DIRECTION`	REVERSED		
`X_POD_OFFSET_MM`	0.0		
`Y_POD_OFFSET_MM`	0.0		

After the test: a coder puts the new numbers into the constants and removes the matching TODO comments; the results go on GitHub issue #6 (and #21 for the offsets). Photos of anything surprising!